THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 11–14: **Exploring Some Geometric Themes, Tales by Dots and Lines, Algebra Play, Area** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(d)** regular tetrahedron	Four faces
2	**(b)** 18	Euler's relation
3	**(c)** 4	Four segments of one-third length
4	**(d)** 45	Arrange in order and take the middle value
5	**(a)** 50	Total = mean × n
6	**(b)** 9.14286	Sum 64 ÷ 7
7	**(a)** $2n$	$2(n+5)-10=2n$
8	**(c)** $8n$	Expand
9	**(c)** 9	Divisibility rule
10	**(d)** 96 cm²	$\frac{1}{2}d_1d_2$
11	**(a)** 120 cm²	$l \times b$
12	**(b)** 8 cm	Area ÷ length

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	$F=5, V=6, E=9$; $5+6-9=2$ (2).	2
14	To display how a quantity changes with time (2).	2
15	$120/5=24$ (2).	2
16	Second row 8, 9; top 17 (2).	2
17	$200000/500=400$ tiles (2).	2

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	$F=7, E=12, V=7$ (2); $F+V-E=2=2$ (1).	2+1
19	Mean $=11.67$ (1); median $=11$ (1); mean (1).	1+1+1
20	Second row 13, 11 (1); top 24 (1); top $=a+2b+c$ (1).	1+1+1
21	Trapezium $=\frac{1}{2}(12+20)\times 8=128$ cm² (2); rectangle 160 cm² (1).	2+1
22	Area 480 m² (1.5); perimeter 94 m, cost ₹ 2820 (1.5).	1½+1½
	OR: Other leg 24 cm (2); area 84 cm² (1).	2+1
23	Number $=3^4=81$ (2); side $=\frac{1}{2^4}$ unit (1).	2+1
	OR: Surface area $=6\times 2^2=24$ cm² (2); 2 cubes (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) Sum $n(n+1)/2$ divided by n (3). (ii) 25.5 (2).	3+2
	OR: (i) Sum ÷ number; middle value (1.5). (ii) Mean 20, median 17 (2). (iii) Median due to the outlier 40 (1.5).	1½+2+1½
25	(i) 216 cm² (2). (ii) 15 cm (1). (iii) $54000/216=250$ tiles (2).	2+1+2
	OR: (i) $\frac{1}{2}\times 20\times(7+9)=160$ cm² (3). (ii) Dividing into two triangles on the diagonal (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 11–14: **Exploring Some Geometric Themes, Tales by Dots and Lines, Algebra Play, Area** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(d)** 7	Count the faces
2	**(c)** 9	Each stage multiplies the count by 3: $3^2 = 9$
3	**(c)** 8	Euler's relation
4	**(d)** line segments	Trend
5	**(a)** no change	Constant value
6	**(c)** bars of heights proportional to values	Category comparison
7	**(d)** even	$n(n+1)$ is a product of consecutive integers
8	**(b)** 3	$3n + 3$
9	**(c)** 52	$4x + 16$
10	**(d)** 84 cm²	$l \times b$
11	**(a)** 6 cm	72 ÷ 12
12	**(d)** 135 cm²	$\frac{1}{2}(a+b)h$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	$3^3 = 27$ (2).	2
14	Ordered: 4, 6, 8, 10, 12, 14; median $(8+10)/2 = 9$ (2).	2
15	$10a + b$ and $10b + a$ (2).	2
16	$2(n+7) - 14 = 2n$ (2).	2
17	96 cm² (2).	2

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	$F = 10$, $E = 24$, $V = 16$ (2); $F + V - E = 2 = 2$ (1).	2+1
19	Mean $= 26$ (1); median $= 26$ (1); both are equal (1).	1+1+1
20	Dates 7, 8, 14, 15 (1); sum 44 (1); algebra check (1).	1+1+1
21	Area 200 m² (1.5); perimeter 66 m, cost ₹ 1980 (1.5).	1½+1½
22	Number $= 3^1 = 3$ (2); side $= \frac{1}{2^1}$ unit (1).	2+1
	OR: Segments multiply by 4 each stage (1); $4^3 = 64$ segments (2).	1+2
23	Mean 140 (1); median 140 (1); comment on spread/outliers (1).	1+1+1
	OR: Total $= 80$ (1); sum of known $= 65$ (1); missing value $= 15$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) A rectangle (3). (ii) A circle (2).	3+2
	OR: (i) Replace the middle third of each side by two segments forming a triangle (3). (ii) Side 1 cm, sides $3 \times 4 \times 4 = 48$ (2).	3+2
25	(i) $3(n+5) - 15 = 3n$, $\div 3 = n$ (2). (ii) Returns the original number (1). (iii) e.g. $(n+5) \times 2 - 2n = 10$ (2).	2+1+2
	OR: (i) $x + (x+1) + (x+7) + (x+8)$ (2). (ii) $4x = 84$, $x = 21$ yes; $4x = 114$ no (3).	2+3

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ANSWER KEY & MARKING SCHEME SET C

Chapters 11–14: **Exploring Some Geometric Themes, Tales by Dots and Lines, Algebra Play, Area** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(d)** 15	Count the edges
2	**(a)** 729	Each stage multiplies the count by 3: $3^6 = 729$
3	**(b)** 1	Plus two circular faces
4	**(a)** mean of the two middle values	Definition
5	**(a)** the total deviation on both sides is equal	Balancing act
6	**(d)** 18	Sum 90 ÷ 5
7	**(c)** $2x + 5$	Translate
8	**(c)** $3n$	$3(n + 8) - 24 = 3n$
9	**(a)** $10 \times 6 + 4$	Place value
10	**(c)** 63 cm²	$\frac{1}{2} \times b \times h$
11	**(a)** 10000 cm²	100 × 100
12	**(a)** equal area	Same base and height

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	5: one square and four triangles (2).	2
14	$22 - 5 = 17$ (2).	2
15	$(n + 1)^2 - n^2 = 2n + 1$ (2).	2
16	Different shapes with equal perimeter can have different areas (1); e.g. 6×2 and 4×4 rectangles (1).	1+1
17	12 cm (2).	2

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Number $= 3^6 = 729$ (2); side $= \frac{1}{2^6}$ unit (1).	2+1
19	Mean $= 46.43$ (1); median $= 45$ (1); mean (1).	1+1+1
20	Dates 4, 5, 11, 12 (1); sum 32 (1); algebra check (1).	1+1+1
21	Area 162 m² (1.5); perimeter 66 m, cost ₹ 1980 (1.5).	1½+1½
22	Total $= 44$ (1); sum of known $= 32$ (1); missing value $= 12$ (1).	1+1+1
	OR: Mean 48.33 (1); median 48.5 (1); comment on spread/outliers (1).	1+1+1
23	Second row 11, 13 (1); top 24 (1); top $= a + 2b + c$ (1).	1+1+1
	OR: Choose equal or closest numbers (1); 18 and 19 (1); product 342 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) Plot months against rainfall and join (2). (ii) 8.33 cm (2). (iii) May to June (1).	2+2+1
	OR: (i) Total 240, remaining 192; mean 24 (3). (ii) Removed numbers average 24 (2).	3+2
25	(i) $1.2 \times 0.8 = 0.96$, a 4% decrease (3). (ii) Changes multiply (2).	3+2
	OR: (i) A diagonal halves a rectangle; or two congruent triangles make a parallelogram (3). (ii) $\frac{1}{2} \times 14 \times 12 = 84$ cm² (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.